The eleven are a narrow subset of the draft-compatible seed positions

Date: 2026-10-24 UTC. **Question:** If the recovered `lrand48` streams are considered under the *contemporaneous* November 1997 X9.62 curve-selection procedure, does its listed mathematics isolate the eleven later named omissions? **Answer:** No. Of the already certified prime-order seed positions in the six prime-field stream prefixes, **69** pass the draft's listed numerical checks: **10 published and 59 unpublished**. The eleven are the subset left by an additional retrospective root/bit-width screen. The 1997 draft does not state that screen. This is a lower bound on qualifying positions under the draft's near-prime procedure, because the inherited census targeted full prime order rather than every possible near-prime cofactor. It is not a historical attempt count.

Source and method

Retrieved primary source: ANSI X9.62 working draft, cover dated 17 November 1997, SHA-256 `0c231d5176fcf0b239b44732213d281b518a0e16cc6f13f7bd4039ec99807b6e`. Section F.3.2 (PDF p. 63, printed p. 55) loops over a verifiably generated curve, counts its order, checks near-primality, MOV and anomalous conditions, and finds a prime-order point. Its notes (PDF p. 64) give `r_min > 2^160`, advise a bound at or below the Hasse lower endpoint, and describe `l_max = 255` as typical. Section F.3.3.b (PDF p. 65) permits arbitrary seed strings of at least 160 bits and arbitrary `a,b` satisfying the specified SHA-1 relation; it does not prescribe either square-root sign or an upper bound on the subgroup-order bit length. The PDF's listed steps also do not specify how a new seed is chosen or which of several accepted examples is published. This description is of the text, not a reconstruction of private software.

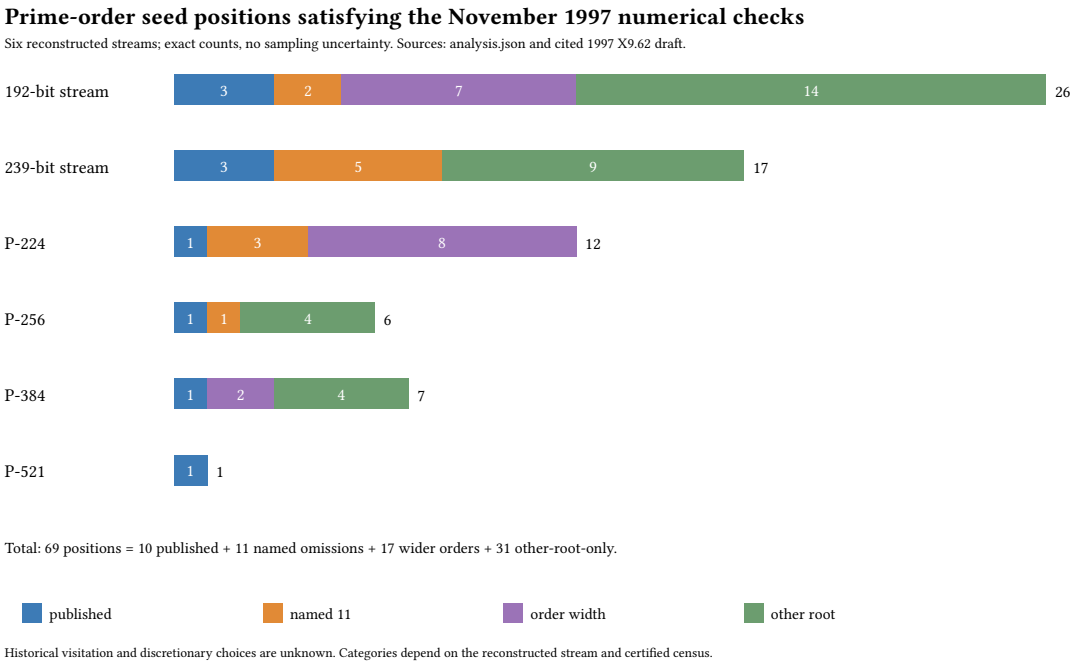
Internal input: the complete prime-order census cross-check and its 82 branch-bound order/point certificates. Their SHA-256 digests, the draft digest, the frozen protocol, and all 69 seed-level rows are retained in analysis.json. The inherited census checked 21,753 stream positions, 11,042 seeded coefficient candidates, and 69 seed positions having at least one certified prime-order root branch. This round reclassified those already certified branches against the dated draft. It did not redo point counting or enumerate all near-prime orders.

Independently checked: analyze.py recomputed the 1997 SHA-1 integer `r` and checked `r b² ≡ (-3)³ (mod p)`, nonsingularity, retained prime-order and point-certificate flags, Hasse membership, `n > 2^160`, `n != p`, and the MOV condition through `B=20` for all 82 branches. It checked a changed-seed control for every branch. A separate bit-string verify.py reproduced all 82 SHA-1 equations and the disjoint 69-position classification without importing the first implementation. A certified prime full order has cofactor one and satisfies the draft's near-primality predicate for any `l_max >= 1`. Since every order lies in its Hasse interval, the draft's advised `r_min` ceiling cannot reject one of these curves. A user-chosen threshold outside that advice or unavailable implementation behavior is a separate historical hypothesis.

Exact partition of the certified set

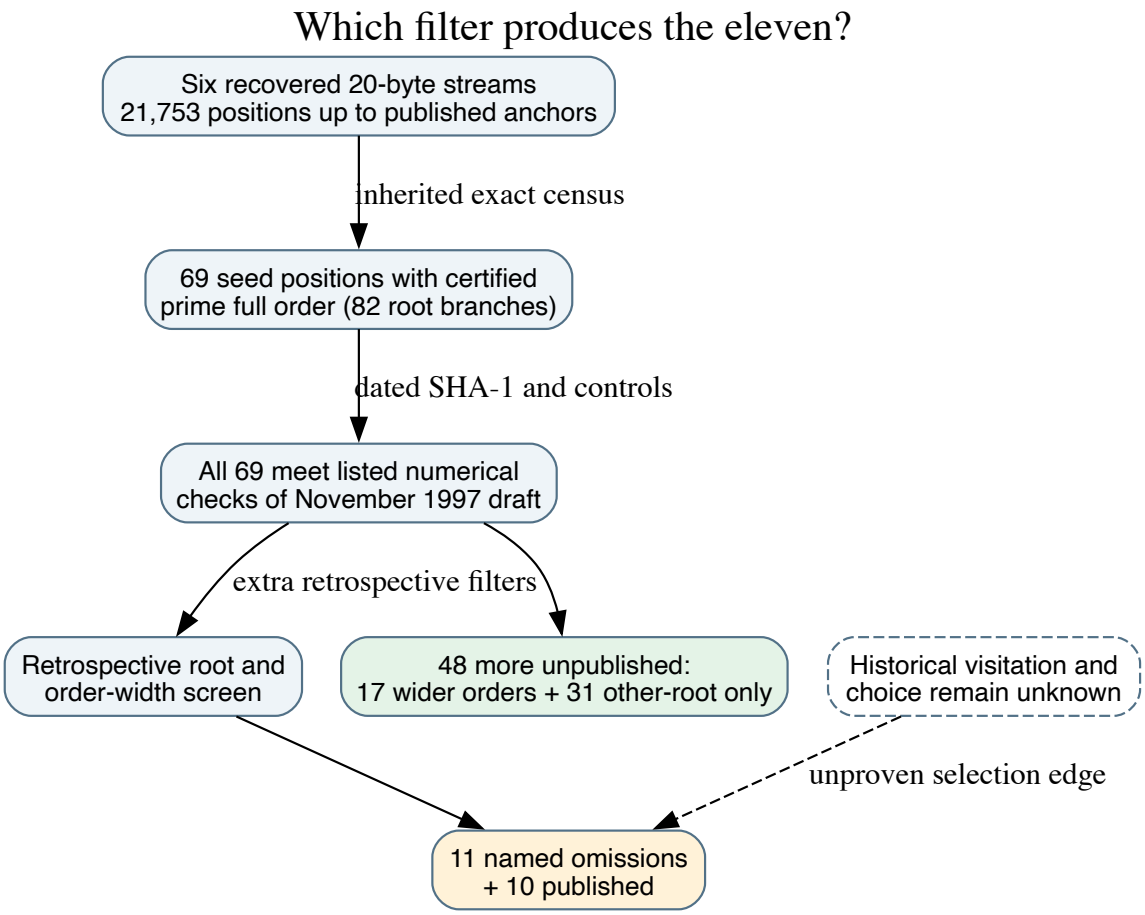
The names `prime192v2` and `prime239v3` below identify shared streams, each containing three different published curves. Every count is a distinct 160-bit seed position; 82 is the number of qualifying root branches, not a second set of positions.

Reconstructed stream	Published	Named eleven	Wider-order only	Other-root only	Draft-compatible certified total
192-bit shared	3	2	7	14	26
239-bit shared	3	5	0	9	17
P-224	1	3	8	0	12
P-256	1	1	0	4	6
P-384	1	0	2	4	7
P-521	1	0	0	0	1
Total	**10**	**11**	**17**	**31**	**69**



Bars count exact seed positions in the reconstructed prefixes. `published`, `named eleven`, `wider-order only`, and `other-root only` are disjoint. Editable chart source. The counts are exact within the inherited prime-order census, so there are no sampling error bars.

The finding is clearest in **P-384**: the familiar eleven-curve list has no P-384 omission, yet six earlier seed positions have a certified prime-order root branch meeting these November 1997 numerical checks. Likewise P-256 has five earlier unpublished certified positions under this broader reading, not just block 876. The P-521 prefix has only its publication among this certified prime-order set; near-prime possibilities were not exhaustively classified here.



Solid arrows are the dated-text and inherited-arithmetic checks. The dashed edge is historical behavior that these records cannot establish. Editable diagram source.

What this says about a possible commonality

Derived: The eleven share a **definition in the retrospective screen**: they are unpublished seed positions whose selected coefficient root has prime order fitting the field's bit width and passes its other tests. That is a useful fixed comparison cohort, but the dated 1997 selection procedure does not isolate it. Any claim that a hidden mathematical defect explains *exactly these eleven* must first say why the other **48** unpublished, draft-compatible prime-order positions were excluded from the alleged historical decision set. One answer might be an actual root convention and an order-width constraint in the original software; neither is documented by the draft or proven by this arithmetic replay. The 1997 draft's freedom to choose a coefficient pair satisfying its hash relation makes the opposite-root cases legitimate numerical possibilities under its text.

Limitation: This does not imply the original program evaluated 59 alternatives, rejected any particular one, or intentionally applied a secret weakness screen. A source-identical generator and run trace could still establish a narrower path. The six prefixes end at their last published position; they do not count other timestamps. The complete near-prime acceptance space may be larger than the 69 full-prime-order positions established here. The point-certificate check uses inherited immutable certificates; the independent verification in this round recomputes the seed equations and partition, not scalar multiplication or point counts. No ECDLP advantage is identified.

Next discriminator: Recover original generator code and candidate-by-candidate logs, including root choice and any subgroup-order bit-length check. A proposed common mathematical weakness should predict labels across the **69** draft-compatible certified positions, or give a source-supported reason to restrict to the eleven, and should include a concrete mechanism and matched published controls. This round checked the existing seed-selection and canonical ECDLP graph sources. No verified security mechanism or transport edge changed, so those graphs remain unchanged; this diagram records the new source-bound selection-space inference.

The unresolved historical rule is tracked in the existing selection-rule problem record. Its exit conditions are not met by this replay.

Reproduction

From the repository root with the hash-matching source PDF at the path fixed in PROTOCOL.md:

```
python3 research/nist_seed_1997_acceptance_20261024/analyze.py
python3 research/nist_seed_1997_acceptance_20261024/verify.py
python3 research/nist_seed_1997_acceptance_20261024/render.py
dot -Tsvg research/nist_seed_1997_acceptance_20261024/inference.dot -o research/
nist_seed_1997_acceptance_20261024/inference.svg
```

The analysis printed `69` positions and `82` branches; the independent verifier printed `verified 82 dated SHA-1 branch equations and 69 disjoint seed-level classes`. Source provenance: X9.62 `retrieved`; census and point certifications `internal`; this round's equations and partition `independently checked`; historical generator behavior `unknown`.